# ASSIGNMENT-4

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Course - BTech CSE

Branch - Cyber Security - A

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Subject - Data Science

Data - 16/4/2026.

Importance:

- Improves data quality and accuracy.
- Makes data ready for analysis
- Helps in better decision-making
- Saves time in analysis
- Handles missing and inconsistent data.

(Q2) Explain string manipulation in python with suitable code examples.

(Ans) String manipulation refers to processing and modifying text data using python functions and methods.

1) Basic String Methods

```
val = 'a,b,c'
# split string
val.split(',')
# ['a', 'b', 'c']
# Remove spaces
places = [x.strip() for x in val.split(',')]
# join strings
':'.join(prices)
```

2) Searching operations

```
val = 'a,b,c'
'c' in val # True
val.find(',') # 1
val.count(',') # 2
```

xplain the concept of Data wrangling and its importance in Data analysis.

Ans) Data wrangling is the process of gathering, collecting, cleaning and transforming raw data into a usable formate for analysis.
It is also called Data Munging.

Steps of Data wrangling:

1) Data Acquisition
collect data from different sources.

2) Data cleaning
Remove errors, missing values.

3) Data Exploration
Understand structure and content

4) Data Transformation
convert into suitable formate.

5) Data Loading
Load into analysis tools.

Key operations:

→ clean
→ Transform
→ Merge
→ Reshape

replace Operation

```
val.replace(',',':')
val.replace(',','')
```

Regular expressions.

```
import re
text = "foo bar\t baz"
re.split('\st', text)
# ['foo', 'bar', 'baz']
```

Pandas Vectorized String Functions

```
import pandas as pd.
import numpy as np.
data = pd.Series({
    'Dave' : 'dave@google.com'
    'Steve' : 'steve@google.com'
})
data.str.contains('google')
```

(03) Explain the general techniques and programming practises used for efficient handling of large datasets in data science

→ Handling large datasets involves efficient techniques and programming practises to manage performance and resources.

Problem with large Data:

→ data quality issue
→ Security and Privacy concern
→ High cost
→ Data integrity issues
→ Difficulty in analysis

Techniques for handling data.

1) Choose the right algorithm.
   > Efficient algorithms improve performance
   > Examples : Online algorithms, Map reduce.

2) Choose the Right Data structure
   ↳ Use appropriate structure like :
   - Tree
   - Hash tables
   - Sparse Data structure.

3) Choose the right Tools.
   ↳ Use optimised tools such as :
   - Numpy
   - Numba
   - Blas
   - Cython

General Programming Tips
→ Do not reinvent the wheel
→ Get maximum use of Hardware
→ Reduce computational needs.

Qn) Explain different data transformation techniques used in preprocessing.

Ans) Data transformation is the process of converting data into a suitable format for analysis.

Techniques :

1) Removing Duplicates.
   df.drop_duplicates()

2) Transforming Data (mapping)
   df['001'] = df['001'].map(lambda x : x*2)

3) Replacing Values
   df.replace('aaa', 0)